

1. Project Description: The Aletia Index

Overview

The **Aletia Index** is a specialized Regulatory Information Management (RIM) and Clinical Trust platform. It serves as a "Seal of Approval" for Artificial Intelligence and Machine Learning-enabled Medical Devices (AI/ML-MD). By bridging the gap between FDA/SAHPRA regulatory clearance and real-world clinical utility, it provides a transparent, data-driven environment for clinicians, NGOs, and manufacturers.

Core Components

- **The Public Index:** A searchable, specialty-categorized database of AI devices.
- **The 10-Point Clinical Assurance Checklist:** A proprietary audit framework covering Clinical Integrity, Regulatory Compliance, Rural Utility, and Continuous Monitoring.
- **The Dual-Timestamp Logic:** A transparency feature ensuring data freshness by separating the "Last Automated Sync" (API data) from the "Last Clinical Review" (Human audit).
- **Manufacturer Portal:** A self-service dashboard for device makers to "claim" their profiles, add marketing data, and update their global regulatory roadmap.

2. Business Plan & Strategy

Value Proposition

- **For Clinicians/NGOs:** Reduces the risk of deploying "black box" AI by providing a clear **Green/Amber/Red Health Status**.
- **For Manufacturers:** Provides a prestigious "Verified" status that accelerates market access and builds trust with institutional buyers.

Market Context (2026)

With over **1,300 FDA-cleared AI devices** and a 49% annual growth rate in submissions, the market is saturated. The Aletia Index capitalizes on the "Trust Deficit" where only <2% of cleared devices currently have randomized clinical trial data.

Operational Part Breakdown

1. **Automated Data Ingestor:** Uses APIs (openFDA, GUDID) to pull raw regulatory facts.
2. **Human-in-the-Loop Audit:** Clinical experts score devices against the 10-Point Checklist.
3. **Revenue Engine:** Integrated payment and subscription management via a user login portal.

3. Revenue Projections (3-Year Forecast)

Calculated in USD (Standard for MedTech) with ZAR equivalents for the South African market.

Revenue Stream	Year 1 (Launch)	Year 2 (Growth)	Year 3 (Scale)
Audit Fees (\$3k/ea)	\$105,000	\$135,000	\$210,000
Verified Subscriptions	\$175,000	\$400,000	\$750,000
API & Reports	\$25,000	\$75,000	\$150,000
Total Revenue (USD)	\$305,000	\$610,000	\$1,110,000
Total Revenue (ZAR)	~R5.5 Million	~R11 Million	~R20 Million

Financial Note: Revenue shifts from labor-intensive audits in Year 1 to high-margin SaaS subscriptions and API licensing by Year 3.

4. Technical Part Breakdown

System Part	Description	Technology Stack (Recommended)
Front-End	Public Index & Manufacturer UI	React / Next.js
Core Database	Storage of device facts & timestamps	PostgreSQL (Reliability & Schema)
API Layer	Connects to openFDA / GUDID	Python (FastAPI or Flask)

Auth & Identity	Secure login for manufacturers	Auth0 or Firebase Auth
Payment Portal	Subscription & Audit billing	Stripe or Peach Payments (SA)
Internal Auditor UI	Workspace for the 10-Point scoring	Internal Admin Dashboard

5. The 10-Point Clinical Assurance Checklist

1. **Peer-Reviewed Validation:** Strength of evidence in clinical literature.
2. **Demographic Accuracy:** Was the AI trained on diverse, relevant populations?
3. **Gold-Standard Comparison:** Performance against the current clinical best practice.
4. **SAHPRA/International Compliance:** Current legal status in target regions.
5. **Adverse Event Protocol:** Is there a clear path for reporting failures?
6. **Cybersecurity & POPIA:** Protection of patient data and system integrity.
7. **Connectivity Resilience:** Does the device work in low-bandwidth (Rural) areas?
8. **Infrastructure Load:** Hardware requirements for local clinics.
9. **Real-World Monitoring:** Performance tracking after the sale.
10. **Change Control Plan:** How the AI updates without breaking safety protocols.